

- (i) **Risk reduction:** Clarification yields strictly lower expected future epistemic risk:

$$\mathbb{E}[\mathcal{R}_{\text{risk}}(b_{t+1}) \mid a_c] < \mathbb{E}[\mathcal{R}_{\text{risk}}(b_{t+1}) \mid a_a].$$

- (ii) **Bounded friction cost:** The clarification cost is finite: $C_{\text{fric}}(a_c) < \infty$.

Then clarification strictly dominates direct answering whenever:

$$\lambda C_{\text{fric}}(a_c) <$$

$$\eta (\mathbb{E}[\mathcal{R}_{\text{risk}}(b_{t+1}) \mid a_a] - \mathbb{E}[\mathcal{R}_{\text{risk}}(b_{t+1}) \mid a_c])$$

where λ and η are the objective weights on interaction cost and epistemic risk, respectively.

Proof sketch. Under the risk-sensitive objective (2), the Q-value of an action decomposes into immediate task reward, weighted epistemic risk, and weighted intervention cost. By condition (i), clarification reduces the expected future risk by some $\Delta_{\text{risk}} > 0$ relative to direct answering. This yields a gain of $\eta \Delta_{\text{risk}}$ in the objective. Clarification also incurs an immediate penalty of $\lambda C_{\text{fric}}(a_c)$. When the former exceeds the latter, the clarificatory action has strictly higher expected return. The dominance condition follows by rearranging terms, and optimality follows from Bellman’s principle of optimality. $\square$

Interpretation. Proposition 1 formalizes the central tradeoff that Frictive Policy Optimization is designed to learn. The ratio η/λ defines an implicit clarification threshold: it specifies how much epistemic risk reduction is required to justify the interaction cost of delaying task completion. When $\eta \gg \lambda$, even modest expected risk reduction warrants clarification; when $\lambda \gg \eta$, only severe uncertainty or hazard justifies intervention.

Crucially, this result identifies what the learning problem actually is. The agent does not need direct access to the latent belief state b_t or the true risk functional $\mathcal{R}_{\text{risk}}$. Instead, the FPO methods (FAR, FPP, GRFR, and FTR) are designed to learn a soft, context-sensitive approximation to this threshold using the tractable friction surrogate $F(h, y)$. In this sense, Proposition 1 provides the normative target that FPO approximates: it characterizes the regime in which clarification is not merely permissible or stylistically cautious, but strictly optimal under risk-sensitive control.

5 The Friction Functional

We now introduce a tractable surrogate for epistemic risk. Since the latent risk functional $\mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t)$ is not directly observable, we approximate it by a structured friction functional defined over dialogue histories and candidate responses.

Before introducing the specific components of the friction functional, it is important to clarify the design principles that govern its construction. The friction functional is not intended as an arbitrary collection of heuristics, nor as an exhaustive catalogue of all possible dialogue failures. Rather, it is a structured surrogate designed to capture a small number of *irreducible epistemic failure modes* that recur across aligned interaction.

We select components according to three criteria. First, each component must correspond to a *distinct epistemic pathology* that is known to induce downstream failure in decision making: miscalibration, inconsistency, latent hazard, silent intent fixing, and unresolved uncertainty (Munn et al., 2024; DeVilling, 2025). Second, each component must be *operationalizable* from observable signals available at training or inference time, without access to the latent belief state or true risk functional. Third, each component must admit a *monotonic interpretation*: increasing values should reliably indicate increasing epistemic danger or normative violation.

Under these criteria, the friction functional is designed to approximate not the full space of dialogue errors, but the minimal set of failure modes that systematically corrupt belief quality over time. **Miscalibration** captures failures of confidence management; **contradiction** captures failures of logical coherence; **hazard** captures failures of normative and safety compliance; **value conflict** captures failures of intent alignment; and **information gain** captures the unique class of interventions that deliberately increase short-term interaction cost in order to reduce long-term epistemic risk.

This design yields a decomposition in which each term corresponds to a theoretically motivated axis of epistemic control, rather than an ad hoc feature. The resulting functional is modular, interpretable, and extensible: additional components may be introduced when new epistemic failure modes are identified, but the present set reflects a minimal basis sufficient to drive risk-sensitive intervention in practice.

These components are chosen to mirror the core intervention classes introduced in Section 3.2, providing a direct mapping from qualitative taxonomy to quantitative control signals.

5.1 Decomposition of Friction

We now make explicit how the friction functional is constructed from a small set of core epistemic failure modes. As argued in Section 3, frictive interventions in dialogue are triggered not by arbitrary surface errors, but by a limited number of recurrent pathologies that systematically degrade belief quality and downstream decision making. These include miscalibration of confidence, logical inconsistency, latent hazard, silent intent fixing, and unresolved uncertainty.

Following the taxonomy of Section 3, we distinguish between **unproductive friction** – epistemic and normative failures that increase downstream risk – and **productive friction** – deliberate interventions that reduce long-term uncertainty at the cost of short-term interaction burden. This distinction is fundamental: unproductive friction should be minimized, while productive friction should be encouraged when epistemic risk is high.

We first define the **unproductive friction** components, each corresponding to a distinct epistemic failure mode:

$$(5) \quad F^-(h, y) = w_1 \text{Unc}(h, y) + w_2 \text{Contr}(h, y) + w_3 \text{Haz}(h, y) + w_4 \text{ValConf}(h, y)$$

This aggregates miscalibration, contradiction, hazard, and value conflict, the four core pathologies that systematically degrade belief quality over time (Gabriel, 2020; Lin et al., 2022; Bai et al., 2022; Ganguli et al., 2022). High values of F^- indicate epistemically dangerous responses that should be avoided.

We then define **productive friction** as the information gain component:

$$(6) \quad F^+(h, y) = w_5 \text{InfoGain}(h, y)$$

This captures interventions that deliberately increase short-term interaction cost in order to reduce long-term epistemic uncertainty. High values of F^+ indicate epistemically beneficial interventions such as clarification and verification.

Finally, we define the **net friction functional** as the difference:

$$(7) \quad F(h, y) = F^+(h, y) - F^-(h, y)$$

This measures the overall epistemic contribution of a response: positive values indicate net epistemic benefit (productive friction dominates), while negative values indicate net epistemic harm (unproductive friction dominates). The net friction functional thus provides a single scalar summary of epistemic quality that can be used directly in reward shaping, preference learning, and trajectory ranking.

This benefit–loss decomposition mirrors standard formulations of net utility in risk-sensitive and cost-sensitive control (Bertsekas, 1995; Sutton and Barto, 2018), and follows the additive decomposition of shaped rewards and auxiliary costs in reinforcement learning (Ng et al., 1999). The key insight is that the five component scores define a low-dimensional epistemic control space: any intervention that degrades belief quality must manifest through at least one axis of F^- , and any beneficial intervention must either reduce F^- or increase F^+ .

Uncertainty and Miscalibration. Following standard miscalibration measures used in expected calibration error for probabilistic models (Niculescu-Mizil and Caruana, 2005; Guo et al., 2017), we define the uncertainty component as:

$$(8) \quad \text{Unc}(h, y) = |\text{conf}(y | h) - \text{acc}(\hat{y}, h)|$$

This measures miscalibration: the absolute difference between the model’s expressed confidence in response y and the estimated probability that y is correct given the dialogue history. High values indicate epistemically dangerous overconfidence or underconfidence. As a component of F^- , miscalibration contributes to unproductive friction and should be minimized.

Contradiction. We penalize explicit contradictions with prior commitments by operationalizing contradiction using standard natural language inference models trained for entailment and contradiction detection (Bowman et al., 2015; Williams et al., 2018).

$$(9) \quad \text{Contr}(h, y) = \max\{0, \text{NLI}_{\text{contr}}(y, h) - \tau_{\text{contr}}\}$$

This assigns friction when the response y contradicts the dialogue history according to a natural language inference model, beyond a tolerance threshold τ_{contr} . It operationalizes logical inconsistency as an epistemic risk signal. As a component of F^- , contradiction contributes to unproductive friction.

Hazard and Normative Violation. We define a hazard component for safety and policy violations, following standard scoring techniques for safety classification and severity-weighted risk as used in aligned language models (Weidinger et al., 2021; Bai et al., 2022).

$$(10) \text{Haz}(h, y) = \mathbf{1}_{\text{haz}}(h, y) \cdot \text{severity}(h, y)$$

This assigns friction proportional to the detected severity of safety, legal, or policy violations triggered by y , making high-risk normative failures dominate the friction score. As a component of F^- , hazard contributes to unproductive friction.

Value Conflict and Silent Intent Fixing. We measure value conflict as a shift in inferred user intent. This is similar to Bayesian belief update and intent tracking in dialogue, where KL divergence is used as a penalty for forced belief revision (Williams and Young, 2007; Traum and Larsson, 2003).

$$(11) \text{ValConf}(h, y) = \text{KL}(P(\iota \mid h) \parallel P(\iota \mid h, y))$$

Specifically, this measures how much committing to response y forces a revision of the inferred user intent distribution. Large KL divergence indicates silent intent fixing, where the model imposes an interpretation rather than resolving ambiguity. As a component of F^- , value conflict contributes to unproductive friction.

Information Gain. We reward epistemically informative interventions as follows (Lindley, 1956; Gervits et al., 2021; Hou et al., 2024):

$$(12) \text{InfoGain}(h, a) = \mathbb{E}_{o \sim P(\cdot \mid h, a)} [H(b_t) - H(b_{t+1})]$$

This measures the expected reduction in belief entropy induced by taking intervention a , rewarding actions that actively reduce epistemic uncertainty. Unlike the preceding components, information gain is *productive*: it captures deliberate epistemic interventions (clarification, verification, information seeking) that trade short-term interaction cost for long-term risk reduction. Accordingly, InfoGain is the sole component of F^+ and enters the net friction functional with a positive sign.

5.2 From Friction to Risk

We now relate friction to latent epistemic risk. The central modeling assumption of this framework

is that *unproductive friction* serves as a tractable surrogate for the unobservable latent epistemic risk:

$$(13) F^-(h, y) \approx \mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t)$$

This follows the standard use of surrogate reward and cost functionals in inverse and preference-based reinforcement learning (Ng and Russell, 2000; Christiano et al., 2017; Rafailov et al., 2023).

Equivalently, the net friction functional approximates the *negation* of epistemic risk, offset by productive friction:

$$(14) F(h, y) \approx F^+(h, y) - \mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t)$$

This relationship has a natural interpretation: responses with high net friction F are those that maximize information gain while minimizing epistemic failure modes—precisely the responses that reduce downstream risk. Consequently, *higher values of F indicate better epistemic quality*, and adding F to a reward signal (as in Friction-Augmented Rewards) has the correct effect of encouraging epistemically beneficial behavior.

Importantly, this approximation is not intended to assert that friction is identical to epistemic risk in any literal or ground-truth sense. Rather, it asserts that a suitably constructed friction functional provides a *behaviorally sufficient proxy* for the aspects of epistemic risk that matter for policy learning. The key requirement is not pointwise accuracy of $F^-(h, y)$ as an estimator of $\mathcal{R}_{\text{risk}}$, but *policy consistency*: actions that reduce latent epistemic risk should, in expectation, also reduce unproductive friction, and vice versa.

More formally, the approximation in (13) is justified under the standard assumption of *surrogate consistency*: there exists a monotone transformation ϕ such that, for all contexts h and admissible actions (a, y) ,

$$(15) \mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t) < \mathcal{R}_{\text{risk}}(b_t, h_t, a'_t, y'_t) \Rightarrow F^-(h, y) < F^-(h, y')$$

That is, the unproductive friction functional need only preserve the ordering of interventions by latent epistemic risk, not their absolute scale. Under this assumption, minimizing expected unproductive friction induces the same set of optimal policies as minimizing the true latent risk, up to equivalence classes of policies that are indistinguishable under the ordering (Ng et al., 1999; Ng and Russell, 2000; Rafailov et al., 2023).